

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2; module 3 restricted to stacks. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which C keyword defines a structure type? (1)
(A) `class`
(B) `union`
(C) `struct`
(D) `enum`
- 2) Object `s` has a nested structure member `admitted` with an integer member `year`. How do you access that `year`? (1)
(A) `s.admitted.year`
(B) `s->admitted.year`
(C) `s.year.admitted`
(D) `s::admitted::year`
- 3) How many non-null characters can a null-terminated string in `char name[40]` hold? (1)
(A) 40
(B) 41
(C) 38
(D) 39
- 4) Which `scanf` format limits a single word to fit in `char name[40]`, including its null terminator? (1)
(A) `%40s`
(B) `%39s`
(C) `%s`
(D) `%41s`
- 5) If `p` points to a structure with integer member `roll`, which expression accesses that member? (1)
(A) `p.roll`
(B) `*p.roll`
(C) `p->roll`
(D) `p::roll`

Section B: Theory questions (25 marks)

- 6) Write a C function that dynamically allocates a contiguous $r \times c$ integer matrix and returns its base pointer, or NULL on failure. Reject nonpositive dimensions. Show how element (i, j) is accessed using pointer arithmetic. Assume the requested size fits in `size_t`. (3)
- 7) Explain a dangling pointer and a memory leak using one short C example of each. State how each error can be prevented. (2)
- 8) A singly linked list stores integer values. Define its node structure and write `void deleteAll(Node **head, int key)` to delete every node whose value equals `key`. Handle an empty list, consecutive matches and matches at the head. Release deleted nodes. (5)
- 9) Write a C function to reverse a singly linked list in place using three pointers. Return the new head and state its time and auxiliary-space complexity. (3)
- 10) A list initially contains $10 \rightarrow 20 \rightarrow 30$. Insert 5 at the head, insert 25 after 20, then delete the tail. Draw the final list and state how many next links must be traversed to locate its tail from the head. (2)
- 11) Convert $A+B*(C-D)^E^F$ to postfix using a stack. Exponentiation has highest precedence and is right-associative. Show the operator stack and output after each input token, then give the final postfix expression. (5)
- 12) Write a C function to check whether a string containing only `(, ), [, and ]` is balanced. Use an array stack of capacity 100 and assume the input has at most 100 characters. Return 1 for balanced and 0 otherwise. (3)
- 13) An editor uses undo and redo stacks. Starting with empty stacks, perform edits A, B and C, then undo twice and redo once. List both stacks from bottom to top. State what happens to the redo stack when a new edit D is made. (2)

End of question paper